

DOWNSTREAM ANALYSIS OF DESEQ2 RESULTS

PATHWAY ANALYSIS

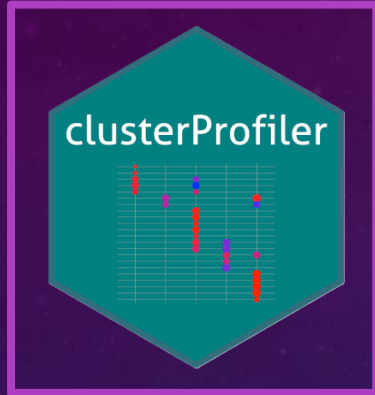
CALEB WATKINS

INBRE 2026 - 06/23/2026

AGENDA

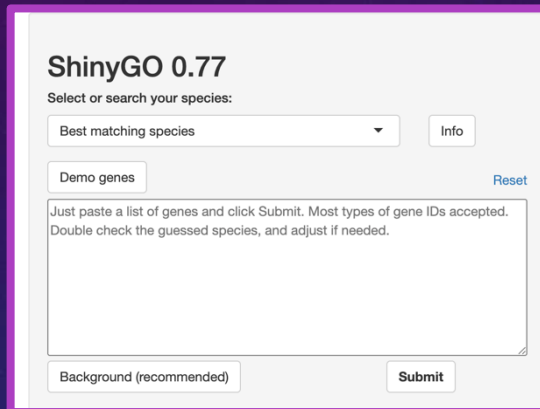
- 01 Overview of three tools for pathway analysis.
- 02 Revisiting DE output from before.
- 03 Taking example lists into ShinyGO.
- 04 IPA demonstration.
- 05 Questions / Help

TOOLS FOR PATHWAY ANALYSIS – CLICK ON A LOGO!



Pathway enrichment analysis right in Rstudio.

- Over-representation analysis (ORA).
- Gene set enrichment analysis (GSEA).
- Can take L2FCs and stats into account.
- Define target reference genome.



Basically, a graphical-user-interface (GUI) for clusterProfiler.

- ORA which uses only the list of genes as input.
- Limited to Ensembl, Ensembl plants, and Ensembl Metazoa genomes.

Click on note to try Shiny v0.85

NOTE:

There is an updated version of ShinyGO (v0.85) but due to high traffic, may not always be available. Use old versions for exploration.



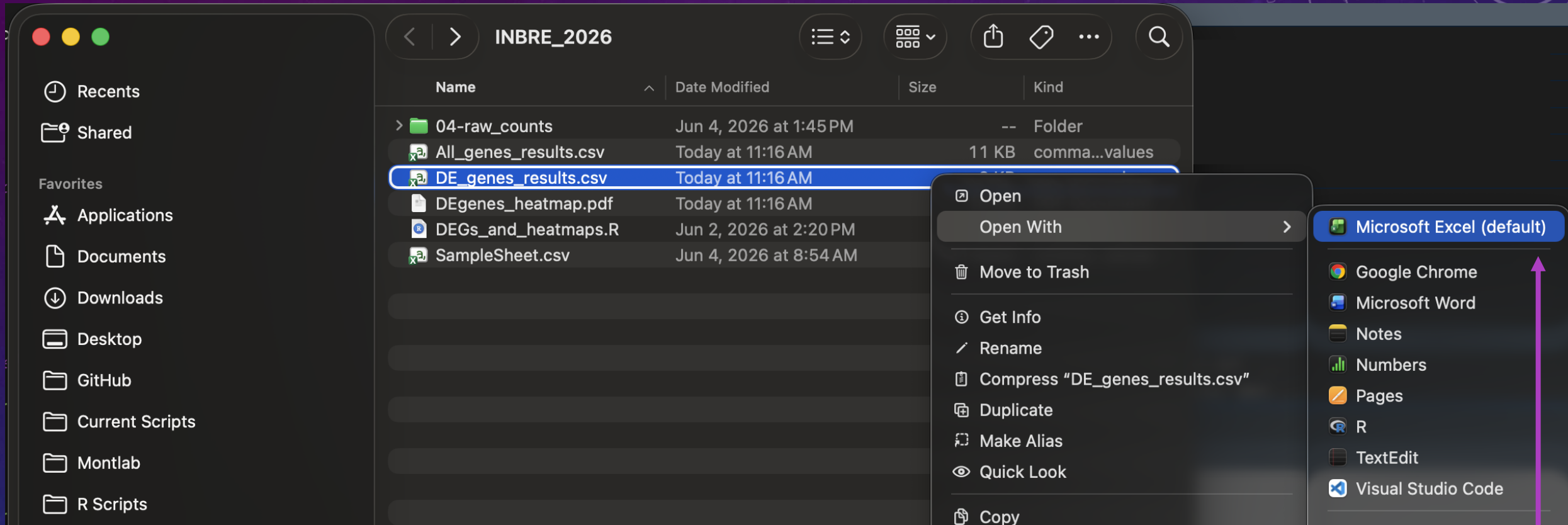
Web-based tool for the analysis, interpretation, and integration of omics data.

- Proprietary and requires a license to access.
- Allows for direct input of DE result for filtering and analysis.
-

A QUICK REVISIT OF OUR DE RESULTS...

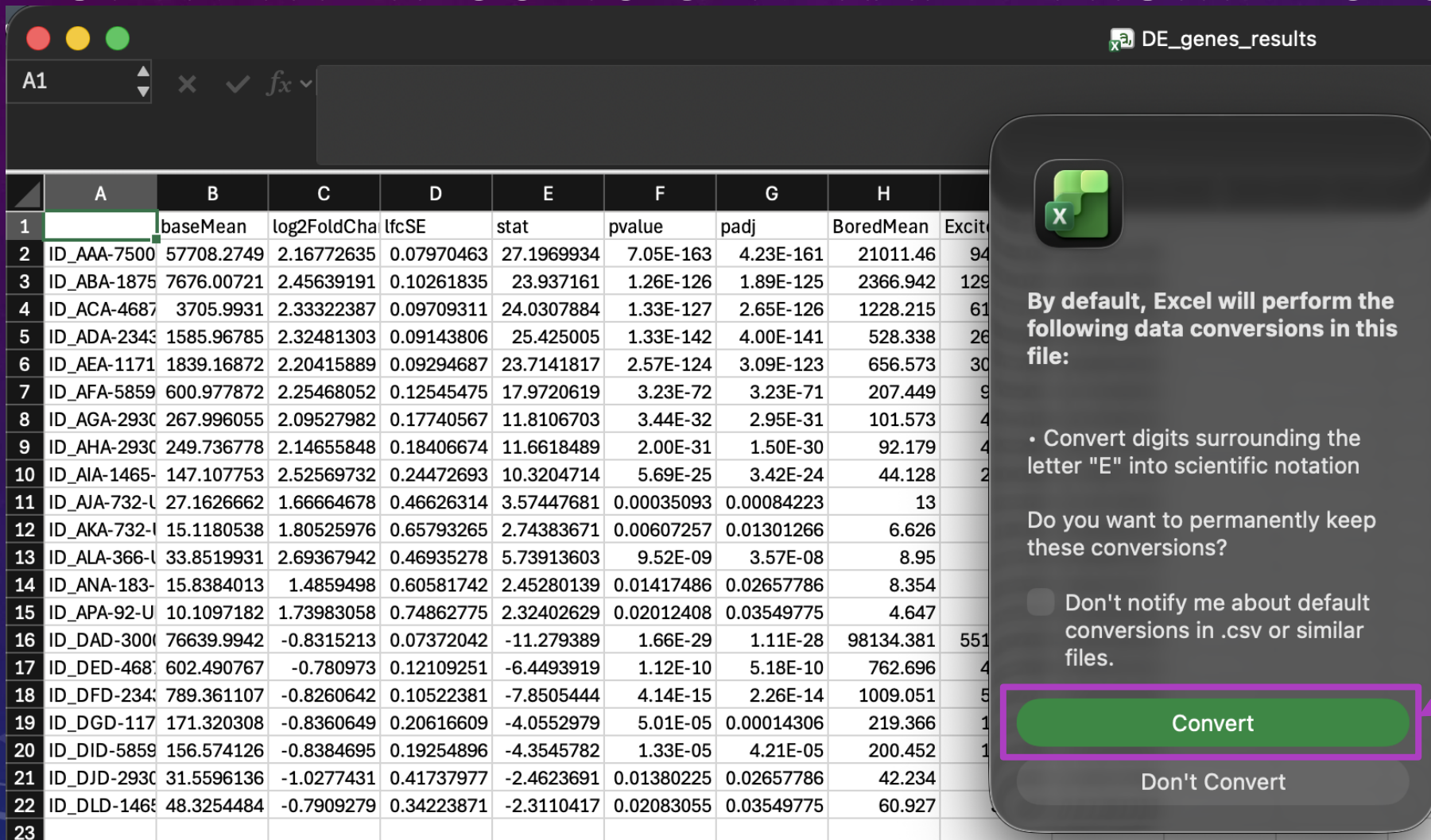
AND HOW TO GET FROM RESULTS TO SHINYGO

OPEN THE RESULTS GENERATED FROM DESEQ2



We want to open this in excel for these next steps.
We could do this in R, but this data is small so excel is sufficient.

OPEN THE RESULTS GENERATED FROM DESEQ2



The image shows an Excel spreadsheet titled "DE_genes_results" with a table of DESeq2 results. The table has columns A through H, with headers: baseMean, log2FoldCha, lfcSE, stat, pvalue, padj, BoredMean, and Excit. The data rows show various gene IDs and their corresponding values. A pop-up dialog box is overlaid on the right side of the spreadsheet, asking if the user wants to convert data to scientific notation. The dialog includes a list of conversions and a checkbox for default conversions. The "Convert" button is highlighted with a red rectangle.

	A	B	C	D	E	F	G	H	
1		baseMean	log2FoldCha	lfcSE	stat	pvalue	padj	BoredMean	Excit
2	ID_AAA-7500	57708.2749	2.16772635	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	94
3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	129
4	ID_ACA-4687	3705.9931	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	61
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	26
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	30
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	9
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	4
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	4
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	2
11	ID_AJA-732-U	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	
12	ID_AKA-732-U	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	
13	ID_ALA-366-L	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	
14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	
16	ID_DAD-300-	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	551
17	ID_DED-468-	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	4
18	ID_DFD-234-	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	5
19	ID_DGD-117	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	1
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	1
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	
22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	
23									

By default, Excel will perform the following data conversions in this file:

- Convert digits surrounding the letter "E" into scientific notation

Do you want to permanently keep these conversions?

☐ Don't notify me about default conversions in .csv or similar files.

Convert

Don't Convert

You may get a pop-up window asking about data conversion.

Select 'Convert' – this is for columns like "padj" to be interpreted as scientific notation instead of character strings, in excel.

LET'S FILTER OUR RESULTS

AutoSave

DE_genes_results

Home Insert Draw Page Layout Formulas Data Review View Automate

Paste Aptos Narrow (Bod... 12 A A

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Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these

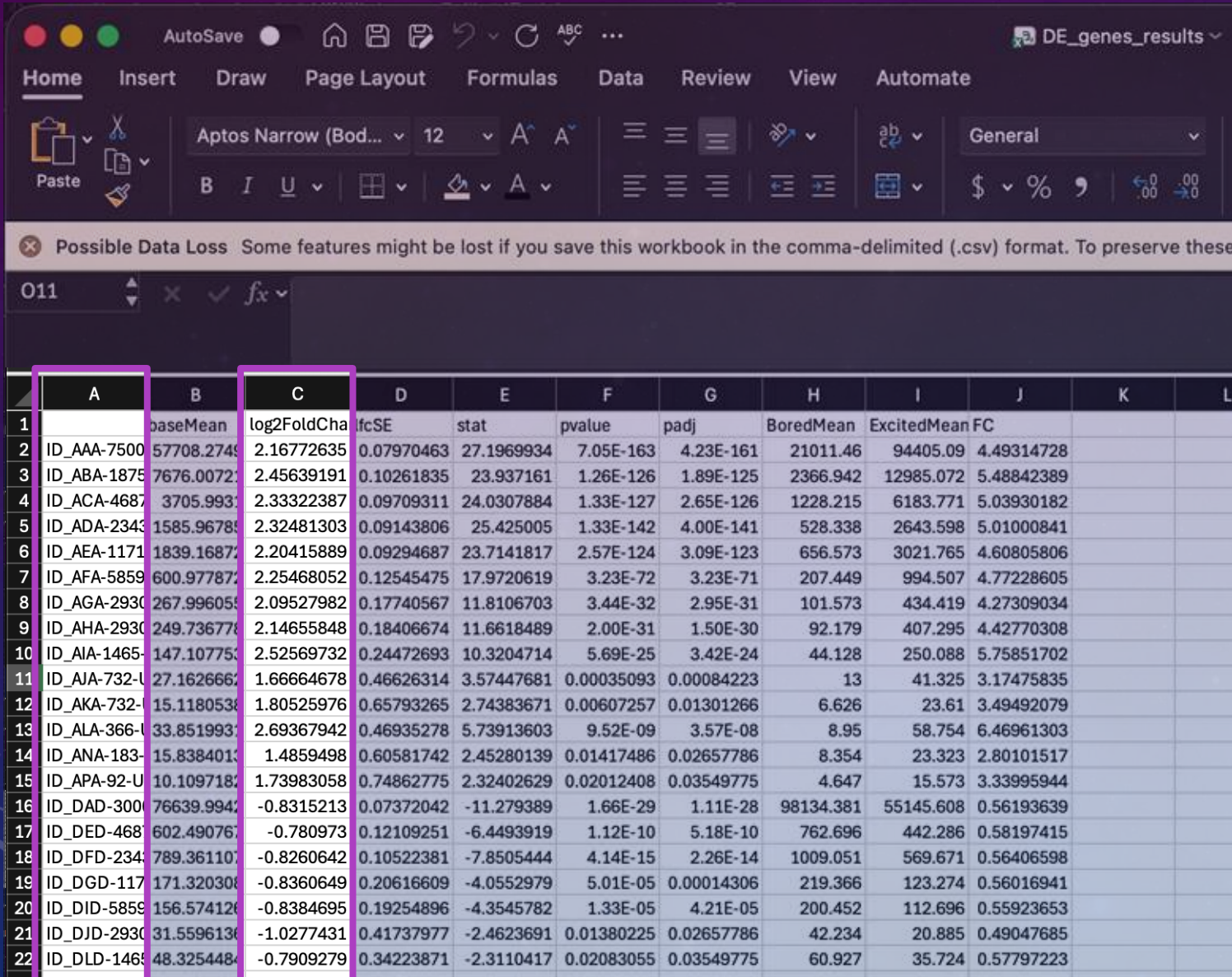
O11

	A	B	C	D	E	F	G	H	I	J	K	L
1		baseMean	log2FoldCha	lfcSE	stat	pvalue	padj	BoredMean	ExcitedMean	FC		
2	ID_AAA-7500	57708.2749	2.16772635	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	94405.09	4.49314728		
3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389		
4	ID_ACA-4687	3705.9931	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182		
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841		
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806		
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	994.507	4.77228605		
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	434.419	4.27309034		
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308		
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702		
11	ID_AJA-732-U	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835		
12	ID_AKA-732-I	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079		
13	ID_ALA-366-U	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303		
14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517		
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944		
16	ID_DAD-300I	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639		
17	ID_DED-468-	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415		
18	ID_DFD-234-	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598		
19	ID_DGD-117	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941		
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653		
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685		
22	ID_DLD-146-	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223		

We are interested in the following columns for our first, quick-glance at pathway enrichment:

- Column A: our gene IDs / names
a.k.a. – rownames
- Column C: "log2FoldChange" (log2FC)
of each gene

LET'S FILTER OUR RESULTS



AutoSave

DE_genes_results

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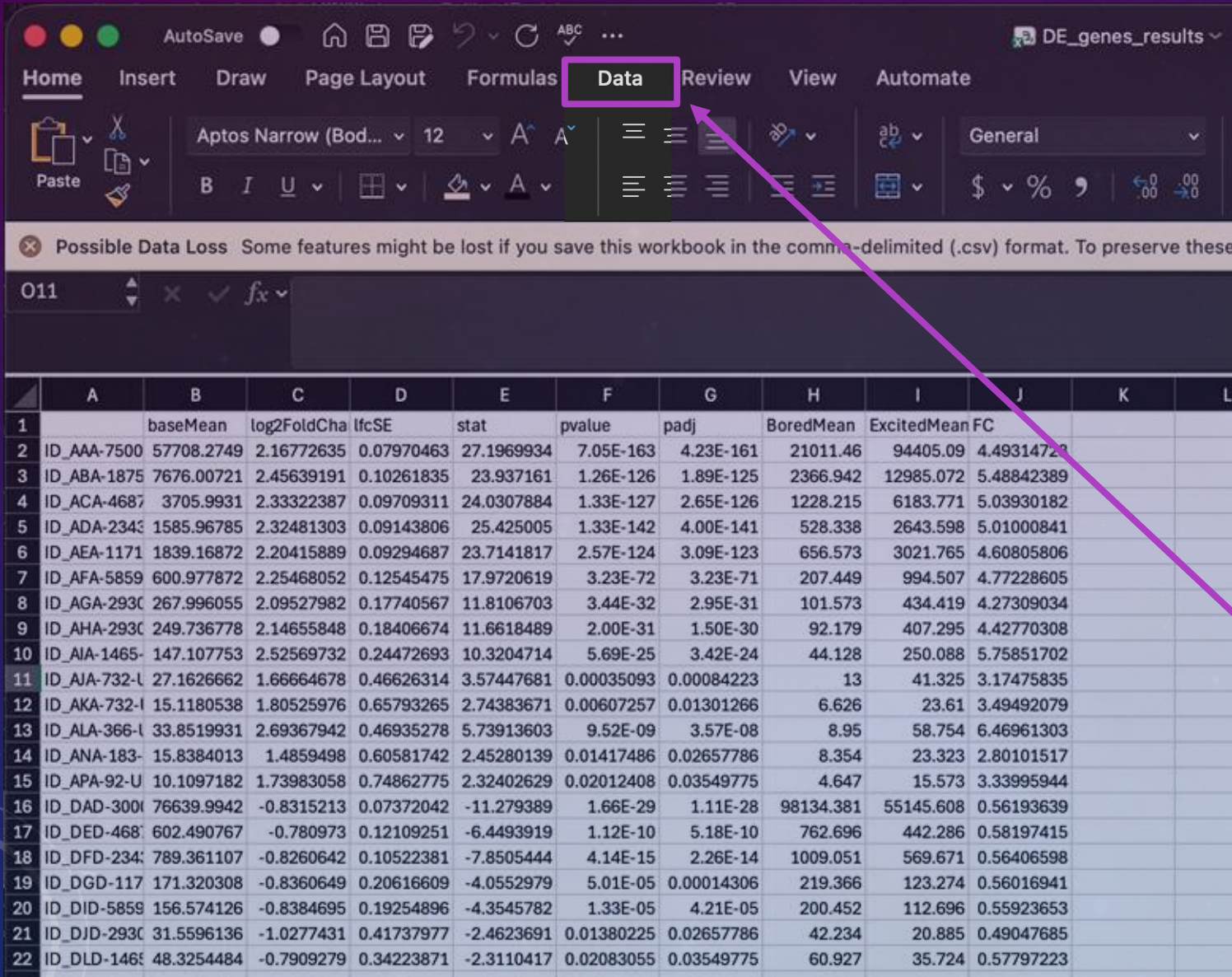
O11

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3	ID_ABA-1875	7676.0072	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389		
4	ID_ACA-4687	3705.993	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182		
5	ID_ADA-2343	1585.9678	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841		
6	ID_AEA-1171	1839.1687	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806		
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9	ID_AHA-2930	249.73677	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308		
10	ID_AIA-1465	147.10775	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702		
11	ID_AJA-732-U	27.162666	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835		
12	ID_AKA-732-U	15.118053	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079		
13	ID_ALA-366-U	33.851993	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303		
14	ID_ANA-183-	15.838401	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517		
15	ID_APA-92-U	10.109718	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944		
16	ID_DAD-300	76639.994	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639		
17	ID_DED-468	602.49076	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415		
18	ID_DFD-234	789.36110	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598		
19	ID_DGD-117	171.32030	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941		
20	ID_DID-5859	156.57412	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653		
21	ID_DJD-2930	31.559613	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685		
22	ID_DLD-146	48.325448	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223		

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DE_genes_results

Home Insert Draw Page Layout Formulas **Data** Review View Automate

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General

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3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389		
4	ID_ACA-4687	3705.9931	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182		
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841		
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806		
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	994.507	4.77228605		
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	434.419	4.27309034		
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308		
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702		
11	ID_AJA-732-L	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835		
12	ID_AKA-732-L	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079		
13	ID_ALA-366-L	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303		
14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517		
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944		
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18	ID_DFD-234-	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598		
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22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223		

We are interested in the following columns for our first, quick-glance at pathway enrichment:

- Column A: our gene IDs / names
a.k.a. – rownames
- Column C: "log2FoldChange" (log2FC)
of each gene

Let's use the 'Data' tab of excel and Filter by the log2FC.

LET'S FILTER OUR RESULTS

The screenshot shows the Microsoft Excel interface with the 'Data' tab selected. The 'Filter' button, represented by a funnel icon, is highlighted with a purple box. A purple arrow points from this button to the column headers of a dataset table. The table has columns A through J, with headers: baseMean, log2FoldChange, lfcSE, stat, pvalue, padj, BoredMean, ExcitedMean, and FC. The data rows contain numerical values for each of these columns.

	A	B	C	D	E	F	G	H	I	J
1		baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	BoredMean	ExcitedMean	FC
2	ID_AAA-7500	57708.2749	2.1677263	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	94405.09	4.49314728
3	ID_ABA-1875	7676.00721	2.4563919	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389
4	ID_ACA-4687	3705.9931	2.3332238	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182
5	ID_ADA-2343	1585.96785	2.3248130	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841
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12	ID_AKA-732-U	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079
13	ID_ALA-366-U	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303
14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944
16	ID_DAD-3000	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639
17	ID_DED-468	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415
18	ID_DFD-234	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598
19	ID_DGD-117	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685
22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223

Highlight columns A : J

We want to highlight all of the columns for our dataset.

From here, click 'Filter'.

You will see dropdown arrows appear on each of the column headers in our dataset.

LET'S FILTER OUR RESULTS

Typically, we would use these dropdown arrows to sort the log2FC between those genes that are upregulated (> 0) or downregulated (< 0).

AutoSave

DE_genes_results

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Stocks Currencies Geography

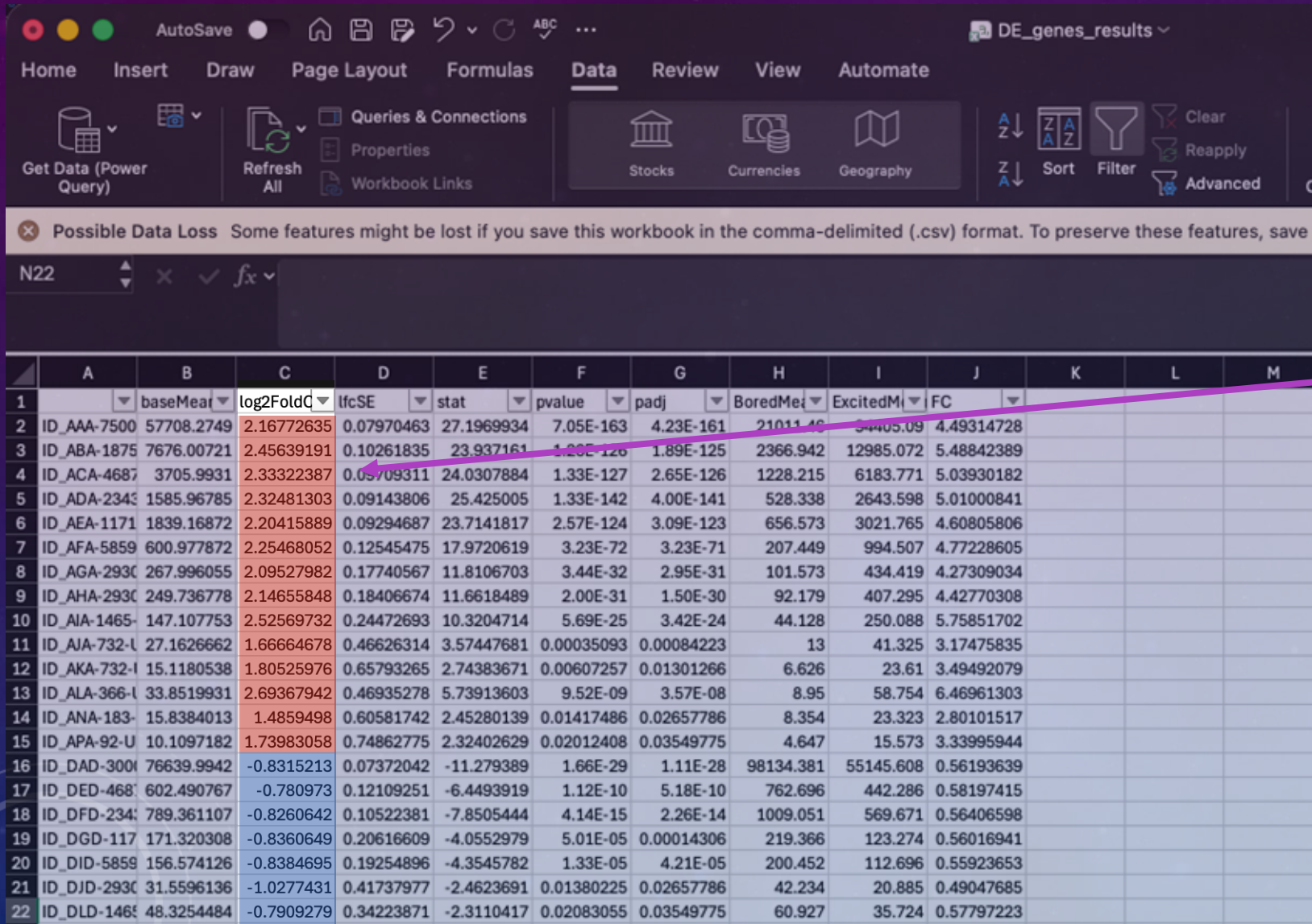
Sort Filter Clear Reapply Advanced

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N22

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	BoredMean	ExcitedMean	FC			
2	ID_AAA-7500	57708.2749	2.16772635	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	94405.09	4.49314728			
3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389			
4	ID_ACA-4687	3705.9931	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182			
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841			
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806			
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	994.507	4.77228605			
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	434.419	4.27309034			
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308			
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702			
11	ID_AJA-732-L	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835			
12	ID_AKA-732-L	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079			
13	ID_ALA-366-L	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303			
14	ID_ANA-183-L	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517			
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944			
16	ID_DAD-300-L	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639			
17	ID_DED-468-L	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415			
18	ID_DFD-234-L	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598			
19	ID_DGD-117-L	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941			
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653			
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685			
22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223			

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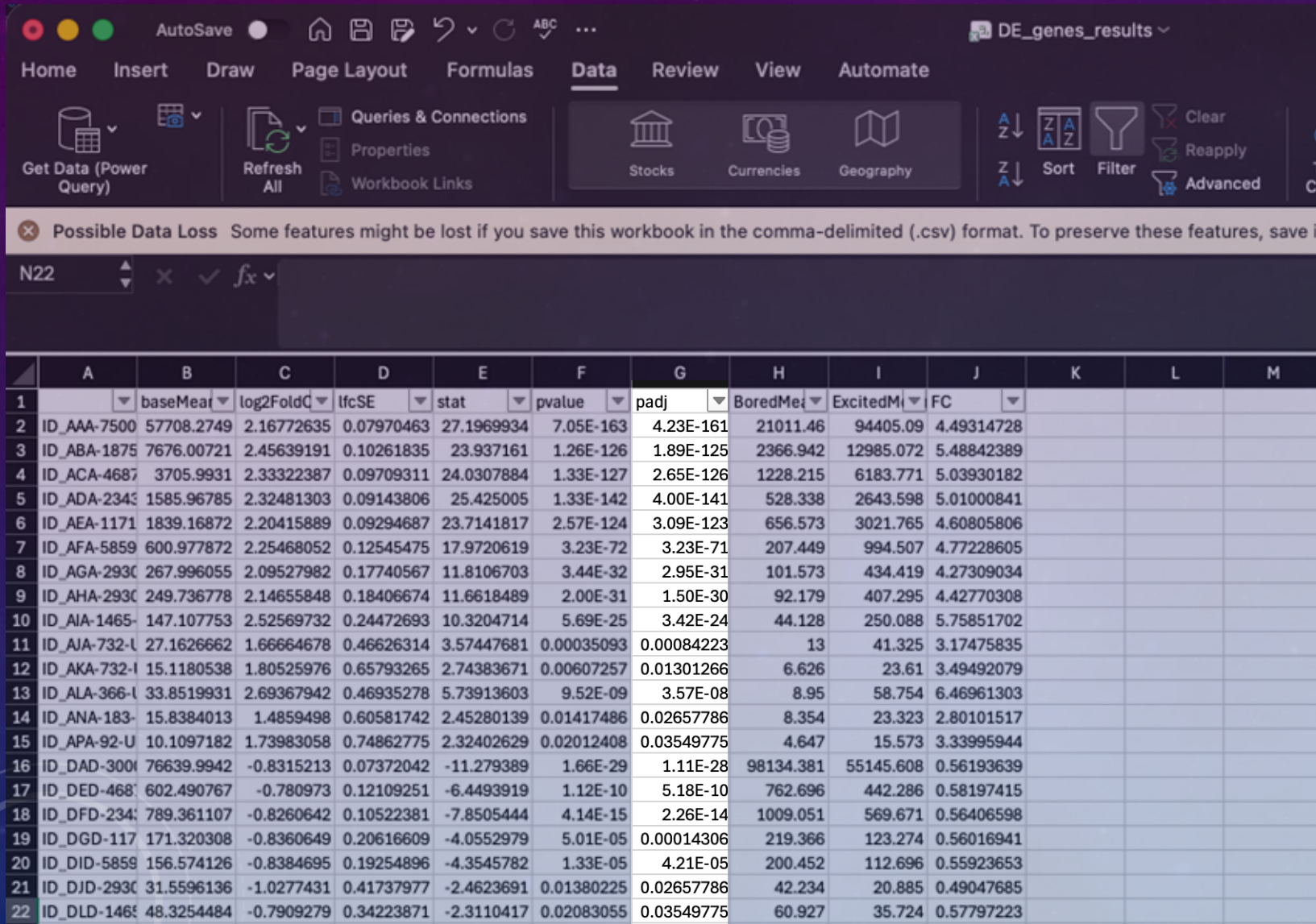
N22

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	BoredMean	ExcitedMean	FC			
2	ID_AAA-7500	57708.2749	2.16772635	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	34405.09	4.49314728			
3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389			
4	ID_ACA-4687	3705.9931	2.33322387	0.05709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182			
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841			
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806			
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	994.507	4.77228605			
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	434.419	4.27309034			
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308			
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702			
11	ID_AJA-732-L	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835			
12	ID_AKA-732-L	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079			
13	ID_ALA-366-L	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303			
14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517			
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944			
16	ID_DAD-300-	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639			
17	ID_DED-468-	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415			
18	ID_DFD-234-	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598			
19	ID_DGD-117	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941			
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653			
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685			
22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223			

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LET'S FILTER OUR RESULTS



DE_genes_results

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N22

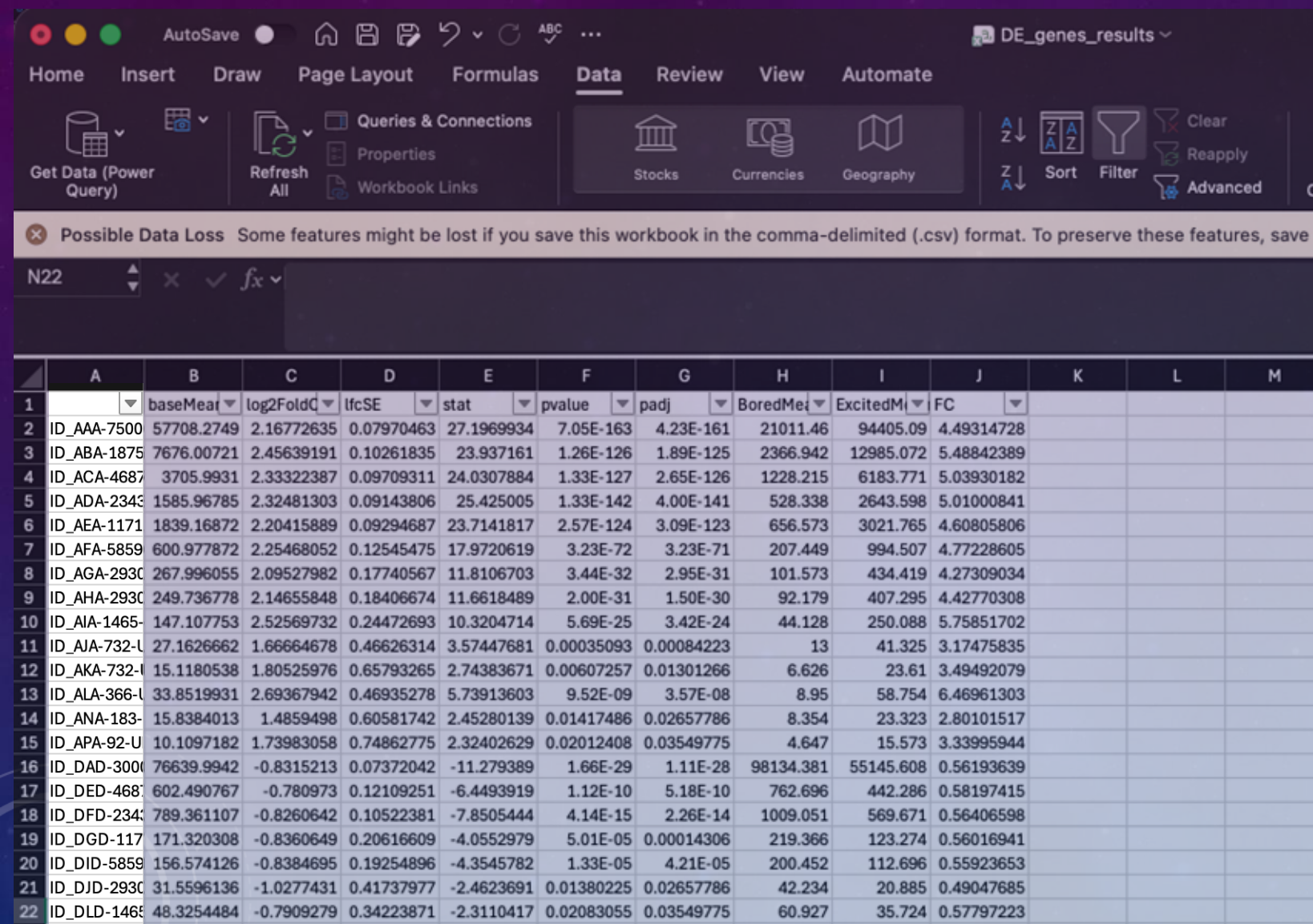
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	BoredMean	ExcitedMean	FC			
2	ID_AAA-7500	57708.2749	2.16772635	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	94405.09	4.49314728			
3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389			
4	ID_ACA-4687	3705.9931	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182			
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841			
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806			
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	994.507	4.77228605			
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	434.419	4.27309034			
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308			
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702			
11	ID_AJA-732-L	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835			
12	ID_AKA-732-L	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079			
13	ID_ALA-366-L	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303			
14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517			
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944			
16	ID_DAD-3001	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639			
17	ID_DED-468-	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415			
18	ID_DFD-234-	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598			
19	ID_DGD-117	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941			
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653			
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685			
22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223			

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Additionally, all our adjusted p-values (padj) are < 0.05 .

LET'S FILTER OUR RESULTS



DE_genes_results

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N22

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	BoredMean	ExcitedMean	FC			
2	ID_AAA-7500	57708.2749	2.16772635	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	94405.09	4.49314728			
3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389			
4	ID_ACA-4687	3705.9931	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182			
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841			
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806			
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	994.507	4.77228605			
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	434.419	4.27309034			
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308			
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702			
11	ID_AJA-732-L	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835			
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14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517			
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944			
16	ID_DAD-3000	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639			
17	ID_DED-4687	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415			
18	ID_DFD-2343	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598			
19	ID_DGD-117	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941			
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653			
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685			
22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223			

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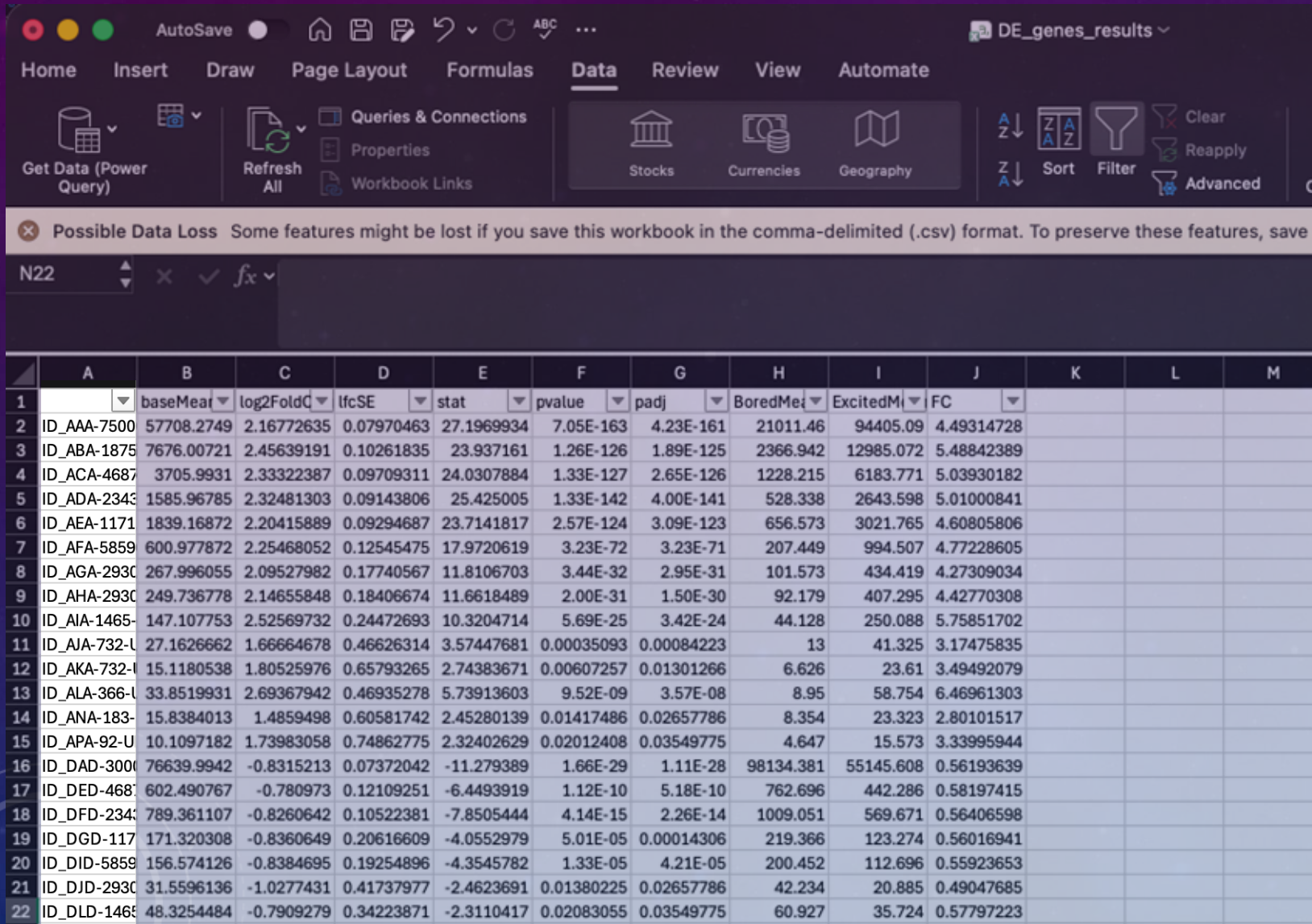
But as you can see. That is already the case here.

Additionally, all our adjusted p-values (padj) are < 0.05 .

Let's copy the gene names from 'Column A' into these lists:

- All
- Upregulated
- Downregulated

LET'S FILTER OUR RESULTS



	A	B	C	D	E	F	G	H	I	J	K	L	M
1		baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	BoredMean	ExcitedMean	FC			
2	ID_AAA-7500	57708.2749	2.16772635	0.07970463	27.1969934	7.05E-163	4.23E-161	21011.46	94405.09	4.49314728			
3	ID_ABA-1875	7676.00721	2.45639191	0.10261835	23.937161	1.26E-126	1.89E-125	2366.942	12985.072	5.48842389			
4	ID_ACA-4687	3705.9931	2.33322387	0.09709311	24.0307884	1.33E-127	2.65E-126	1228.215	6183.771	5.03930182			
5	ID_ADA-2343	1585.96785	2.32481303	0.09143806	25.425005	1.33E-142	4.00E-141	528.338	2643.598	5.01000841			
6	ID_AEA-1171	1839.16872	2.20415889	0.09294687	23.7141817	2.57E-124	3.09E-123	656.573	3021.765	4.60805806			
7	ID_AFA-5859	600.977872	2.25468052	0.12545475	17.9720619	3.23E-72	3.23E-71	207.449	994.507	4.77228605			
8	ID_AGA-2930	267.996055	2.09527982	0.17740567	11.8106703	3.44E-32	2.95E-31	101.573	434.419	4.27309034			
9	ID_AHA-2930	249.736778	2.14655848	0.18406674	11.6618489	2.00E-31	1.50E-30	92.179	407.295	4.42770308			
10	ID_AIA-1465	147.107753	2.52569732	0.24472693	10.3204714	5.69E-25	3.42E-24	44.128	250.088	5.75851702			
11	ID_AJA-732-L	27.1626662	1.66664678	0.46626314	3.57447681	0.00035093	0.00084223	13	41.325	3.17475835			
12	ID_AKA-732-L	15.1180538	1.80525976	0.65793265	2.74383671	0.00607257	0.01301266	6.626	23.61	3.49492079			
13	ID_ALA-366-L	33.8519931	2.69367942	0.46935278	5.73913603	9.52E-09	3.57E-08	8.95	58.754	6.46961303			
14	ID_ANA-183-	15.8384013	1.4859498	0.60581742	2.45280139	0.01417486	0.02657786	8.354	23.323	2.80101517			
15	ID_APA-92-U	10.1097182	1.73983058	0.74862775	2.32402629	0.02012408	0.03549775	4.647	15.573	3.33995944			
16	ID_DAD-3000	76639.9942	-0.8315213	0.07372042	-11.279389	1.66E-29	1.11E-28	98134.381	55145.608	0.56193639			
17	ID_DED-4687	602.490767	-0.780973	0.12109251	-6.4493919	1.12E-10	5.18E-10	762.696	442.286	0.58197415			
18	ID_DFD-2343	789.361107	-0.8260642	0.10522381	-7.8505444	4.14E-15	2.26E-14	1009.051	569.671	0.56406598			
19	ID_DGD-117	171.320308	-0.8360649	0.20616609	-4.0552979	5.01E-05	0.00014306	219.366	123.274	0.56016941			
20	ID_DID-5859	156.574126	-0.8384695	0.19254896	-4.3545782	1.33E-05	4.21E-05	200.452	112.696	0.55923653			
21	ID_DJD-2930	31.5596136	-1.0277431	0.41737977	-2.4623691	0.01380225	0.02657786	42.234	20.885	0.49047685			
22	ID_DLD-1465	48.3254484	-0.7909279	0.34223871	-2.3110417	0.02083055	0.03549775	60.927	35.724	0.57797223			

Let's copy the gene names from 'Column A' into these lists:

- All
- Upregulated
- Downregulated

All

ID_AAA-750000-UP-4
ID_ABA-187500-UP-4
ID_ACA-46875-UP-4
ID_ADA-23438-UP-4
ID_AEA-11719-UP-4
ID_AFA-5859-UP-4
ID_AGA-2930-UP-4
ID_AHA-2930-UP-4
ID_AIA-1465-UP-4
ID_AJA-732-UP-4
ID_AKA-732-UP-4
ID_ALA-366-UP-4
ID_ANA-183-UP-4
ID_APA-92-UP-4
ID_DAD-3000000-DOWN-0.5
ID_DED-46875-DOWN-0.5
ID_DFD-23438-DOWN-0.5
ID_DGD-11719-DOWN-0.5
ID_DID-5859-DOWN-0.5
ID_DJD-2930-DOWN-0.5
ID_DLD-1465-DOWN-0.5

UP

ID_AAA-750000-UP-4
ID_ABA-187500-UP-4
ID_ACA-46875-UP-4
ID_ADA-23438-UP-4
ID_AEA-11719-UP-4
ID_AFA-5859-UP-4
ID_AGA-2930-UP-4
ID_AHA-2930-UP-4
ID_AIA-1465-UP-4
ID_AJA-732-UP-4
ID_AKA-732-UP-4
ID_ALA-366-UP-4
ID_ANA-183-UP-4
ID_APA-92-UP-4

DOWN

ID_DAD-3000000-DOWN-0.5
ID_DED-46875-DOWN-0.5
ID_DFD-23438-DOWN-0.5
ID_DGD-11719-DOWN-0.5
ID_DID-5859-DOWN-0.5
ID_DJD-2930-DOWN-0.5
ID_DLD-1465-DOWN-0.5

We can take lists of genes into gene set enrichment tools like ShinyGO. 

EXAMPLE GENE LISTS FOR ENRICHMENT

ALL

TNF
 IL1B
 IL6
 CXCL8
 CXCL10
 CCL2
 CCL5
 IFNG
 STAT1
 STAT2
 IRF1
 IRF7
 NFKB1
 RELA
 ICAM1
 VCAM1
 PTGS2
 TLR2
 TLR4
 MYD88
 CD80
 CD86
 HLA-DRA
 HLA-DRB1
 B2M
 TAP1
 TAP2
 GBP1
 GBP5
 ISG15
 MX1
 OAS1
 IFI44L
 IFIT1
 IFIT3
 CCR7
 SELL
 FOXP3
 IL10
 TGFB1
 MRC1
 CD163
 ARG1
 PPARG
 KLF4

UP

TNF
 IL1B
 IL6
 CXCL8
 CXCL10
 CCL2
 CCL5
 IFNG
 STAT1
 STAT2
 IRF1
 IRF7
 NFKB1
 RELA
 ICAM1
 VCAM1
 PTGS2
 TLR2
 TLR4
 MYD88
 CD80
 CD86
 HLA-DRA
 HLA-DRB1
 B2M
 TAP1
 TAP2
 GBP1
 GBP5
 ISG15
 MX1
 OAS1
 IFI44L
 IFIT1
 IFIT3

DOWN

CCR7
 SELL
 FOXP3
 IL10
 TGFB1
 MRC1
 CD163
 ARG1
 PPARG
 KLF4

ShinyGO 0.85.1
Human

Select by FDR, sort by Fold Enrichment ▼

Enrichment FDR	nGenes	Pathway Genes	Fold Enrichment	Pathways (click for details)
4.1E-24	13	49	137.1	Malaria
1.8E-24	14	65	111.3	Inflammatory bowel disease
1.2E-23	14	76	95.2	Leishmaniasis
2.5E-24	15	93	83.4	Rheumatoid arthritis
4.1E-25	16	108	76.6	Toll-like receptor signaling pathway
8.5E-22	14	102	70.9	Chagas disease
1.8E-15	10	77	67.1	Pertussis
6.9E-18	12	102	60.8	Amoebiasis
2.7E-29	20	171	60.4	Influenza A
3.4E-16	11	99	57.4	AGE-RAGE signaling pathway in diabetic complications
1.8E-17	12	111	55.9	Toxoplasmosis
4.8E-27	19	181	54.3	Herpes simplex virus 1 infection
4.1E-17	12	119	52.1	TNF signaling pathway
1.8E-24	18	202	46.1	Epstein-Barr virus infection
8.5E-22	16	183	45.2	NOD-like receptor signaling pathway
2.7E-16	12	139	44.6	Measles
4.0E-20	15	179	43.3	Tuberculosis
1.7E-15	12	163	38	Hepatitis B
5.7E-19	15	215	36.1	Lipid and atherosclerosis
4.0E-20	16	234	35.3	Coronavirus disease-COVID-19

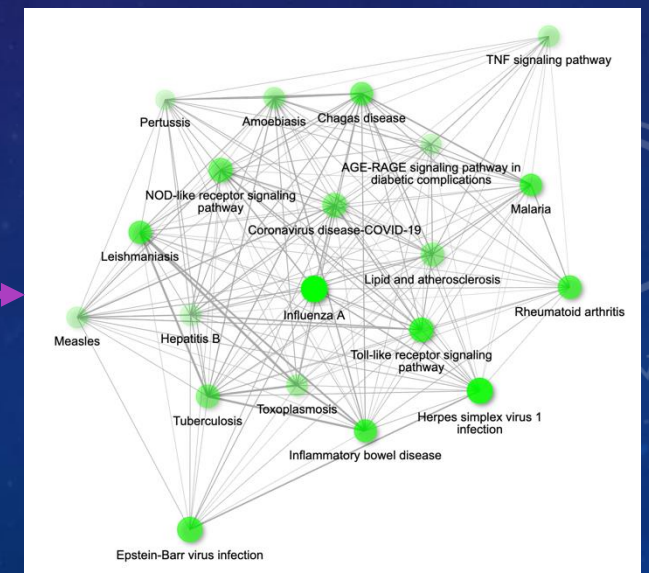
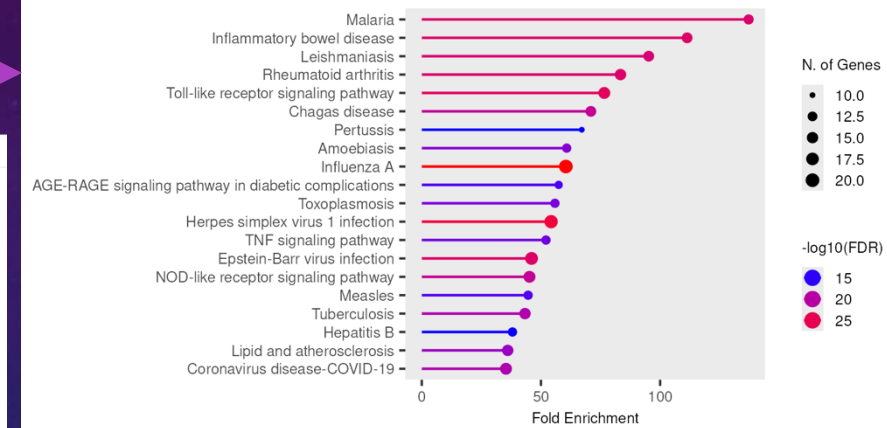
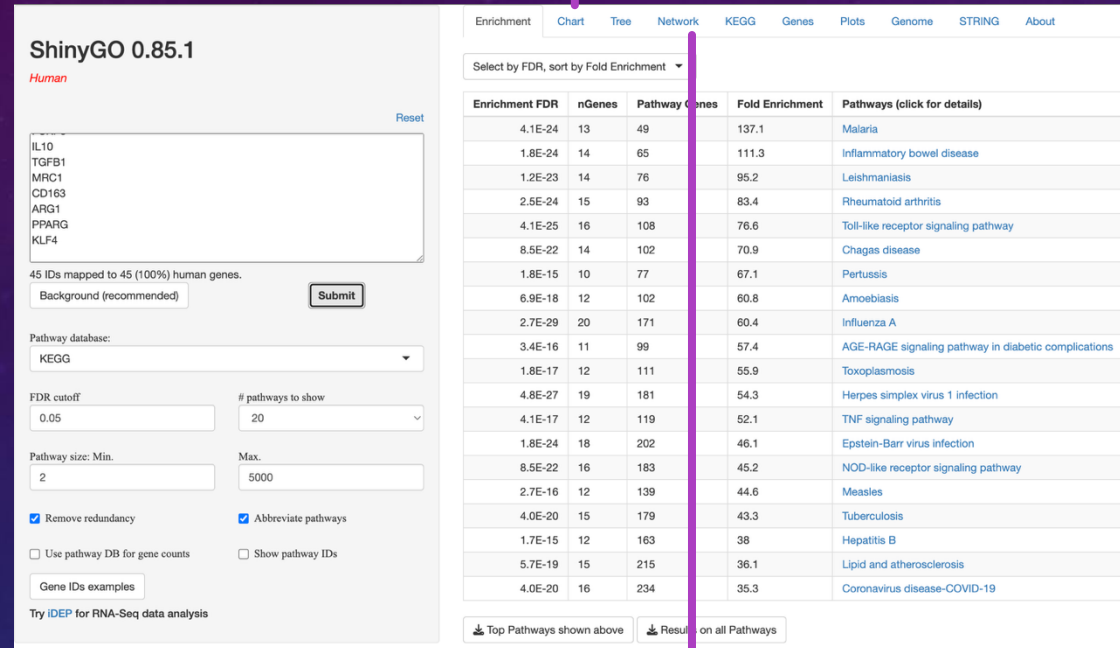
☒ Remove redundancy ☒ Abbreviate pathways
☐ Use pathway DB for gene counts ☐ Show pathway IDs
 Gene IDs examples
 Try [iDEP](#) for RNA-Seq data analysis

Top Pathways shown above Results on all Pathways

EXAMPLE GENE LISTS FOR ENRICHMENT

ALL

TNF	HLA-DRB1
IL1B	B2M
IL6	TAP1
CXCL8	TAP2
CXCL10	GBP1
CCL2	GBP5
CCL5	ISG15
IFNG	MX1
STAT1	OAS1
STAT2	IFI44L
IRF1	IFIT1
IRF7	IFIT3
NFKB1	CCR7
RELA	SELL
ICAM1	FOXP3
VCAM1	IL10
PTGS2	TGFB1
TLR2	MRC1
TLR4	CD163
MYD88	ARG1
CD80	PPARG
CD86	KLF4
HLA-DRA	



NOTE: You can't determine, based on your data here, what is activated or suppressed without the L2FCs!

EXAMPLE GENE LISTS FOR ENRICHMENT

DOWN

CCR7
SELL
FOXP3
IL10
TGFB1
MRC1
CD163
ARG1
PPARG
KLF4

ShinyGO 0.85.1

Human

IL10
TGFB1
MRC1
CD163
ARG1
PPARG
KLF4

10 IDs mapped to 10 (100%) human genes.

Background (recommended)

Submit

Pathway database:

KEGG

FDR cutoff

0.05

pathways to show

20

Pathway size: Min.

2

Max.

5000

☒ Remove redundancy

☒ Abbreviate pathways

☐ Use pathway DB for gene counts

☐ Show pathway IDs

Enrichment

Chart

Tree

Network

KEGG

Genes

Plots

Genome

STRING

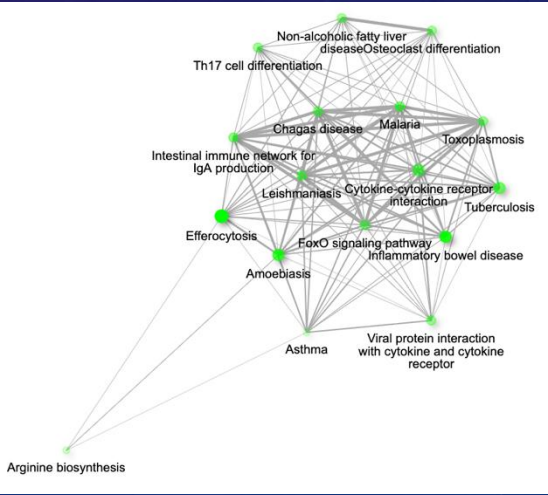
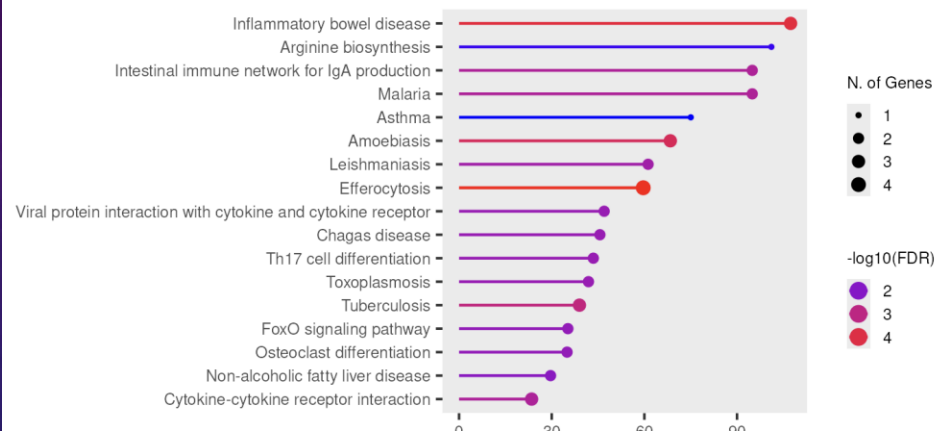
About

Select by FDR, sort by Fold Enrichment

Enrichment FDR	nGenes	Pathway Genes	Fold Enrichment	Pathways (click for details)
7.9E-05	3	65	107.3	Inflammatory bowel disease
3.9E-02	1	23	101.1	Arginine biosynthesis
2.1E-03	2	49	94.9	Intestinal immune network for IgA production
2.1E-03	2	49	94.9	Malaria
5.0E-02	1	31	75	Asthma
2.1E-04	3	102	68.4	Amoebiasis
3.7E-03	2	76	61.2	Leishmaniasis
2.5E-05	4	156	59.6	Efferocytosis
5.3E-03	2	99	47	Viral protein interaction with cytokine and cytokine receptor
5.3E-03	2	102	45.6	Chagas disease
5.3E-03	2	107	43.5	Th17 cell differentiation
5.3E-03	2	111	41.9	Toxoplasmosis
8.3E-04	3	179	39	Tuberculosis
6.5E-03	2	132	35.2	FoxO signaling pathway
6.5E-03	2	133	35	Osteoclast differentiation
8.4E-03	2	157	29.6	Non-alcoholic fatty liver disease
2.1E-03	3	297	23.5	Cytokine-cytokine receptor interaction

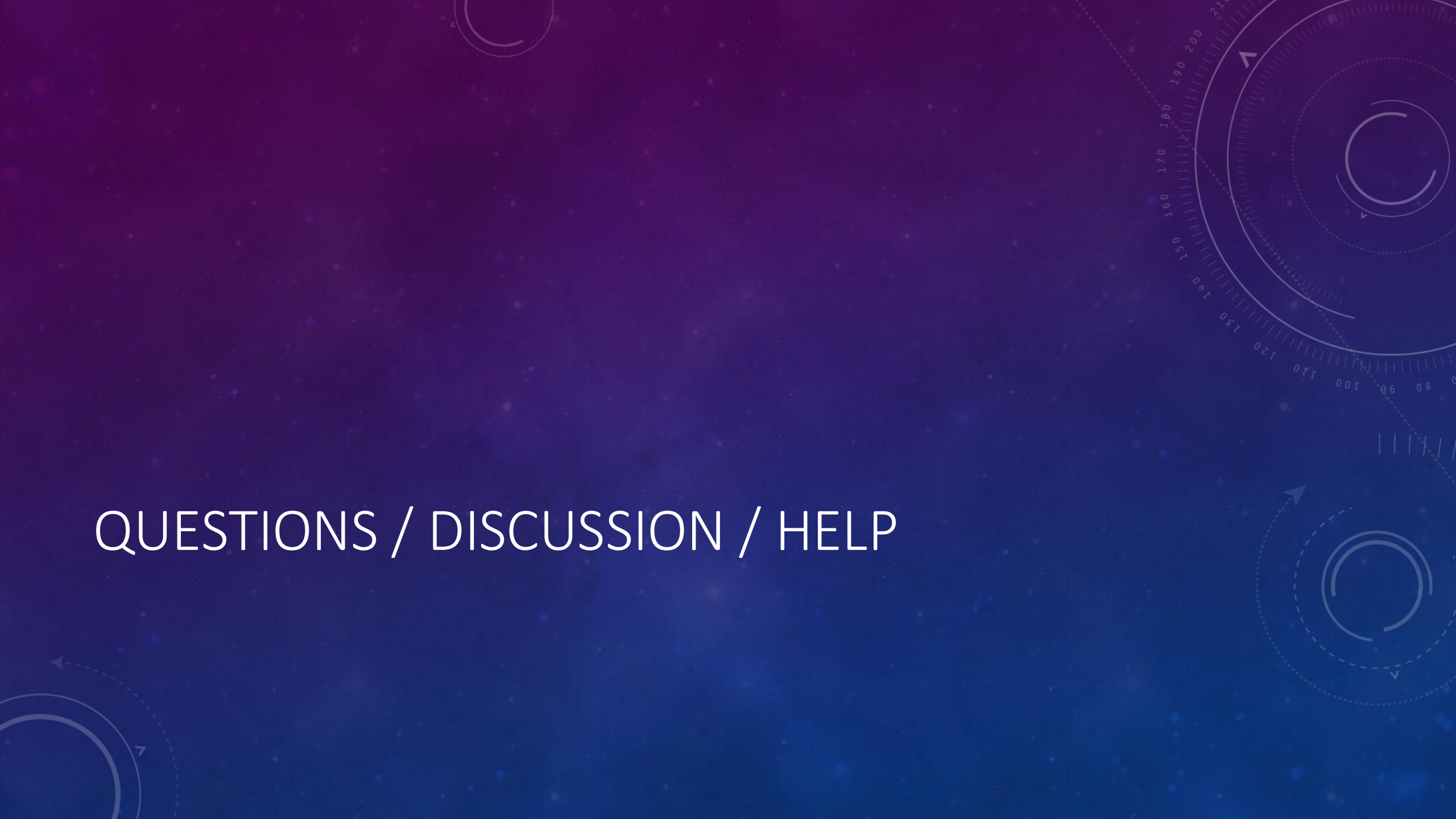
Top Pathways shown above

Results on all Pathways



NOW A DETOUR TO EXPLORE IPA





QUESTIONS / DISCUSSION / HELP